

Surname / *Van*:

Initials / *Voorletters*:

Student number / *Studentenommer*:

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FSK 116

Semester Test 1 - *Semestertoets 1*

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

Date/*Datum*: 04/03/2016

Time/*Tyd*: 2.5 hours/*ure*

Total / *Totaal*: 70

Examiner/*Eksaminator*: RQ Odendaal

Moderator/*Moderator*: Dr JM Nel

Test register: Tear carefully on the line above and hand to the invigilator.

Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 116: Semester Test 1 / *Semestertoets 1*

04/03/2016

Surname/ *Van*:

Initials/ *Voorletters*:

Student no./ *Studenteno.*:

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Signature/ *Handtekening*:

Venue/ *Lokaal*:

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1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*
2. No pencil work will be marked. Do not use “tippex”. *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof (“tippex”) gebruik nie.*
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. Numeriese waardes mag slegs in die laaste stap van ’n berekening vervang word! ’n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.*
6. **Units must be used when numbers are substituted else marks will be subtracted. Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.**
7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in ’n ernstige lig beskou word.*

Constants and formulas / Konstantes en formules

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$; $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$; $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6 \text{ m}$

1 Question 1 / Vraag 1

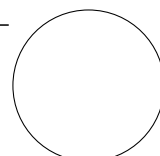
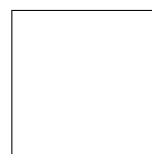
[7]

Andy Warhol was a famous artist. He once said that “everyone will be world-famous for 15 minutes”. *Andy Warhol was ’n befaamde kunstenaar. Hy het gesê “elke persoon sal vir 15 minute wêreld-bekend wees”.*

- (a) If we define a new unit of 1 warhol = 15 minutes, how long is 1 megawarhol in years? Show the conversion factors in your calculation.

As ons definieer dat 1 warhol = 15 minute, hoe lank is 1 megawarhol in jare? Toon die omskakelingsfaktore in jou berekening.

(4)



- (b) Photons move at a constant speed of 300 000 km/s. How far (in meters) would a photon move in 1 Megawarhol? Show the conversion factors in your calculation. Use the given data in your calculation.

Fotone beweeg teen 'n konstante spoed van 300 000 km/s. Hoe ver (in meters) sal 'n foton beweeg in 1 Megawarhol? Toon die omskakelingsfaktore in jou berekening. Gebruik die gegewe data in jou berekening.

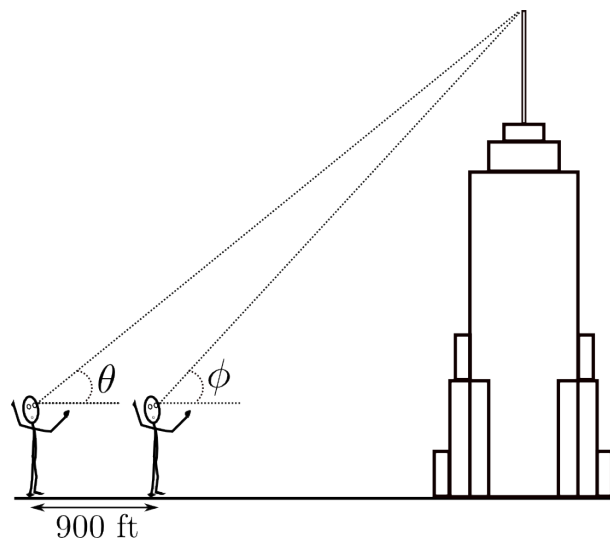
(3)

2 Question 2 / Vraag 2

[6]

During a visit to New York, a tourist decides to estimate the height of the Empire State building. She measures the angle θ of elevation of the spire atop the building as 20° . After walking 900 feet (ft) closer to the building, she measures a new angle of elevation as $\phi = 25^\circ$. Use this data to **estimate** the height of the Empire State building. You can express your answer in feet. Use an annotated (labeled) sketch to explain your calculation. The sketch below is not to scale.

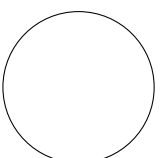
*Gedurende 'n besoek aan New York besluit 'n toeris om die hoogte van die Empire State-gebou te bepaal. Sy meet die hoogtehoek θ van die toring bo-op die gebou as 20° . Nadat sy 900 voet (ft) nader aan die gebou geloop het, meet sy 'n nuwe hoogtehoek van $\phi = 25^\circ$. Gebruik dié data om die hoogte van die Empire State-gebou **benaderd te bepaal**. Jy kan jou antwoord in voet laat. Gebruik 'n skets met aantekeninge om jou berekening te verduidelik. Die figuur hieronder is nie volgens skaal geteken nie.*



3

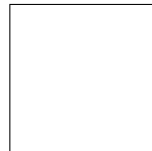
[8]

Die figuur toon die snelheidsgrafiek van 'n trein wat **begin by die oorsprong** by $t = 0$ s.



- (a) Find the acceleration of the train at $t = 3.0$ s.
Vind die versnelling van die trein by $t = 3.0$ s.

(2)



- (b) Draw the corresponding position and acceleration graphs for the train. Your graphs must be neat, and to scale.

Teken die ooreenstemmende posisie- en versnellingsgrafieke vir die trein. Jou grafieke moet netjies wees, en tot skaal geteken word.

(6)



4

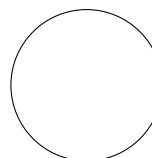
Question 4 / Vraag 4

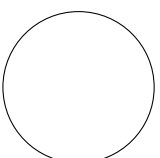
[5]



A particle is confined to move along the x -axis of a coordinate system with **constant** acceleration. If the position of the particle is given by $x = x_i + v_{xi}t + \frac{1}{2}a_xt^2$ at any time t , and the velocity of the particle is given as $v_x = v_{xi} + a_xt$ for any time t , prove that $v_x^2 = v_{xi}^2 + 2a_x\Delta x$. You must use the two given equations in your derivation.

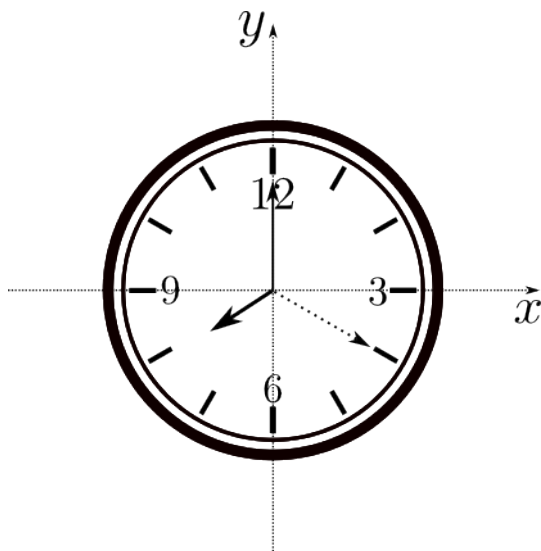
*'n Deeltjie is beperk om te beweeg langs die x -as van 'n koördinaat-stelsel met **konstante** versnelling. Indien die posisie van die deeltjie gegee word deur $x = x_i + v_{xi}t + \frac{1}{2}a_xt^2$ vir enige tyd t , en die snelheid van die deeltjie word gegee deur $v_x = v_{xi} + a_xt$ vir enige tyd t , bewys dat vir enige tyd t is $v_x^2 = v_{xi}^2 + 2a_x\Delta x$. Jy moet die twee gegewe vergelykings in jou afleiding gebruik.*

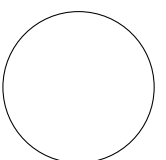




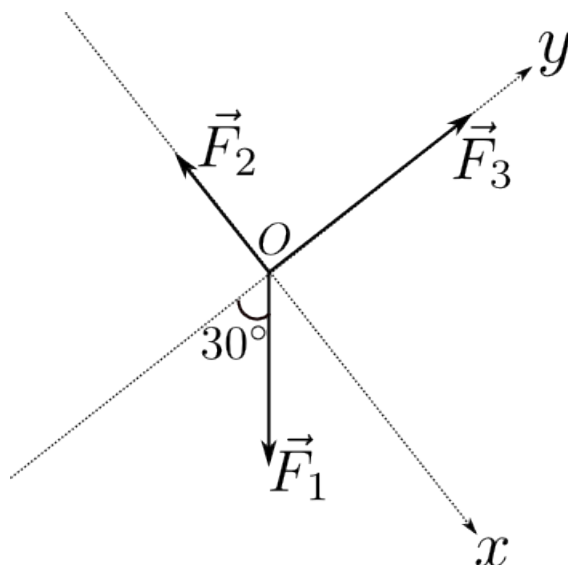
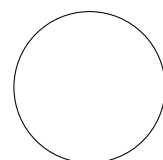
This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[9]

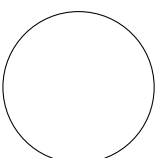


[illegible]

Drie vektore word in die figuur getoon, maar is nie volgens skaal geteken nie. Die som van die drie vektore is $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$. Indien $R_y = 0$ en $F_2 = 0.5 F_3$, vind R_x in terme van F_1 .

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[7]

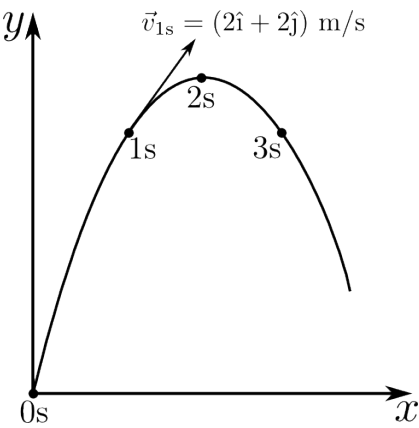
[illegible]

9

Question 9 / Vraag 9

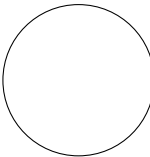
[14]

A physics student on Planet Exidor throws a ball, and it follows the parabolic path as shown in the figure. The ball's position is shown at 1 s intervals until $t = 3$ s. At $t = 1$ s, the ball's velocity is $\vec{v}_{1s} = (2\hat{i} + 2\hat{j})$ m/s.
'n Fisika-student op Planeet Exidor gooi 'n bal, en dit volg die paraboliese pad soos aangetoon in die figuur. Die bal se posisie word gewys vir 1 s intervalle tot by $t = 3$ s. By $t = 1$ s, is die bal se snelheid $\vec{v}_{1s} = (2\hat{i} + 2\hat{j})$ m/s.



- (a) Determine the ball's velocity, in vector notation, at $t = 2$ s and $t = 3$ s.
Bepaal die bal se snelheid, in vektor notasie, by $t = 2$ s en $t = 3$ s.

(4)



- (b) What is the value of g on Planet Exidor?
Wat is die waarde van g op Planeet Exidor?

(4)

- (c) Determine the ball's velocity at $t = 0$ s.
Bepaal die bal se snelheid by $t = 0$ s.

(3)

- (d) Determine the ball's launch angle.
Bepaal die bal se lanseer-hoek.

(3)

END OF TEST / EINDE VAN TOETS

